Gothenburg patient study

rifampicin (all results)

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TRANSLATIONAL IMAGING IN DRUG SAFETY ASSESSMENT

Contents

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Two-scan results

1.1. Data summary

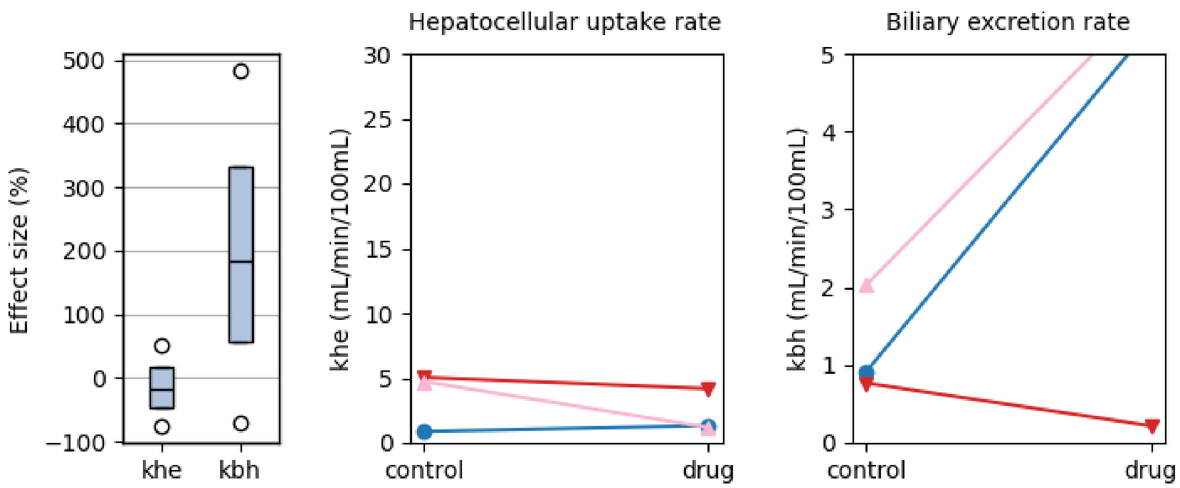

Figure 1.1: Effect size (%) on hepatocellular uptake (k_he, left) and biliary excretion (k_bh, right) of gadoxetate. The boxplot shows median, interquartile range and 95 percent range. The line plots show individual values for hepatocellular uptake (k_he, middle) and biliary excretion (k_bh, right) of gadoxetate of the control (left of plot) and treatment (right of plot). Grey lines are healthy controls with rifampicin injection.

parameter	count	mean	std	min	25%	50%	75%	max
khe effect size (%)	3.0	-13.5	63.2	-74.7	-45.9	-17.1	17.2	51.4
khe control (mL/min/100cm3)	3.0	3.56	2.32	0.88	2.81	4.74	4.9	5.05
khe drug (mL/min/100cm3)	3.0	2.24	1.69	1.2	1.27	1.34	2.76	4.19
kbh effect size (%)	3.0	197.7	276.8	-71.3	55.8	182.8	332.3	481.7
kbh control (mL/min/100cm3)	3.0	1.24	0.69	0.77	0.84	0.92	1.48	2.03
kbh drug (mL/min/100cm3)	3.0	3.77	3.08	0.22	2.78	5.34	5.55	5.75

Table 1.1: Effect size and absolute values of hepatocellular uptake (k_he) and biliary excretion (k_bh) of gadoxetate

1.2. Liver biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for Cl (0-35min)	0.22215	1.0	3.9
AUC for Cl (0-inf)	0.64061	0.53	1.82
Biliary excretion rate	0.24472	0.94	0.13
Biliary tissue excretion rate	0.2924	0.83	0.2
Final biliary excretion rate	0.10562	1.61	0.01
Final hepatocellular uptake rate	0.71655	0.5	1.81
Gut dispersion	0.3646	0.72	3.27
Gut mean transit time	0.40617	0.68	0.29
Hepatocellular mean transit time	0.60347	0.54	0.43
Hepatocellular tissue uptake rate	0.9425	0.47	1.11
Hepatocellular uptake rate	0.37905	0.71	3.25
Initial biliary excretion rate	0.7832	0.49	0.68
Initial hepatocellular uptake rate	0.24645	0.93	6.13
Liver T1-MOLLI at 45min	0.98716	0.47	1.01
Liver T1-MOLLI at baseline	0.07944	1.93	3.61
Liver T1-MOLLI at scan 2	0.54424	0.57	0.74
Liver extracellular volume fraction	0.01003	6.37	14.81
Liver plasma clearance	0.41662	0.67	2.16
RE for R1l at 20min	0.21596	1.01	7.28
RE for Sl at 20min	0.16283	1.22	14.35

Table 1.2: Results of a pairwise comparison testing for differences in liver biomarkers between control and treatment. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	60.0 (26.0)	45.8 (16.0)	-19.9 (26.0)
AUC for Cl (0-inf)	mM*sec	257.0 (320.0)	189.0 (78.0)	22.9 (81.0)
Biliary excretion rate	mL/min/100cm3	1.24 (0.78)	3.77 (3.5)	198.0 (310.0)
Biliary tissue excretion rate	/min	0.0255 (0.019)	0.0579 (0.056)	140.0 (260.0)
Final biliary excretion rate	mL/min/100cm3	1.84 (2.6)	5.8 (0.47)	735.0 (700.0)
Final hepatocellular uptake rate	mL/min/100cm3	4.1 (3.5)	3.15 (3.2)	20.4 (100.0)
Gut dispersion	%	81.3 (8.5)	76.1 (9.5)	-6.18 (11.0)
Gut mean transit time	sec	48.5 (12.0)	54.3 (5.6)	15.7 (29.0)
Hepatocellular mean transit time	min	49.7 (27.0)	103.0 (180.0)	81.6 (290.0)
Hepatocellular tissue uptake rate	/min	0.123 (0.079)	0.119 (0.087)	31.2 (120.0)
Hepatocellular uptake rate	mL/min/100cm3	3.56 (2.6)	2.24 (1.9)	-13.5 (71.0)
Initial biliary excretion rate	mL/min/100cm3	3.1 (3.0)	3.71 (3.5)	77.0 (260.0)
Initial hepatocellular uptake rate	mL/min/100cm3	3.01 (2.6)	1.34 (0.63)	-39.3 (45.0)
Liver T1-MOLLI at 45min	sec	0.793 (0.14)	0.793 (0.13)	0.33 (10.0)
Liver T1-MOLLI at baseline	sec	0.983 (0.12)	0.916 (0.089)	-6.57 (3.5)
Liver T1-MOLLI at scan 2	sec	0.807 (0.16)	0.827 (0.11)	3.25 (8.1)
Liver extracellular volume fraction	mL/100cm3	26.9 (6.0)	19.0 (6.0)	-30.1 (8.6)
Liver plasma clearance	L/min	0.0397 (0.033)	0.0277 (0.031)	-7.69 (76.0)
RE for R1l at 20min	%	22.5 (11.0)	14.5 (4.8)	-28.2 (35.0)
RE for Sl at 20min	%	19.4 (8.3)	11.1 (3.3)	-36.9 (32.0)

Table 1.3: Mean values along with their 95 percent confidence intervals for all liver biomarkers of the control and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.3. Systemic biomarkers

Biomarker	p-value	Bayes Factor	Odds Ratio
AUC for Cb (0-35min)	0.19167	1.1	0.19
AUC for Cb (0-inf)	0.40364	0.68	0.16
Body extraction fraction	0.88032	0.47	1.41
Cardiac output	0.11571	1.52	60.19
Heart-lung dispersion	0.26587	0.88	0.29
Heart-lung mean transit time	0.94794	0.47	0.92
Organs blood mean transit time	0.19056	1.1	0.07
Organs extraction fraction	0.14116	1.34	61.81
Organs extravascular mean transit time	0.83565	0.48	0.64
RE for R1b at 20min	0.22259	0.99	0.19
RE for Sb at 20min	0.38367	0.7	0.4

Table 1.4: Results of a pairwise comparison testing for differences in systemic biomarkers between control and treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	45.2 (12.0)	62.6 (28.0)	35.6 (34.0)
AUC for Cb (0-inf)	mM*sec	88.4 (49.0)	187.0 (150.0)	181.0 (250.0)
Body extraction fraction	%	4.54 (3.7)	3.76 (5.4)	220.0 (590.0)
Cardiac output	L/min	6.96 (0.62)	5.75 (0.6)	-17.1 (12.0)
Heart-lung dispersion	%	38.1 (11.0)	43.6 (7.2)	17.9 (26.0)
Heart-lung mean transit time	sec	19.9 (6.0)	20.1 (2.6)	5.76 (33.0)
Organs blood mean transit time	sec	29.5 (3.9)	34.5 (3.8)	18.2 (20.0)
Organs extraction fraction	%	19.8 (1.8)	17.2 (0.46)	-12.7 (10.0)
Organs extravascular mean transit time	min	6.43 (5.3)	7.36 (2.8)	66.4 (130.0)
RE for R1b at 20min	%	20.0 (6.3)	29.4 (15.0)	44.3 (53.0)
RE for Sb at 20min	%	20.4 (7.9)	26.9 (19.0)	24.5 (40.0)

Table 1.5: Mean values along with their 95 percent confidence intervals for all systemic biomarkers at the control and and treatment visit. The last column shows the relative change at the treatment visit. The results are ranked by their p-value, with most significant differences at the top of the list.

1.4. Case notes

1.4.1. Subject GOT-001

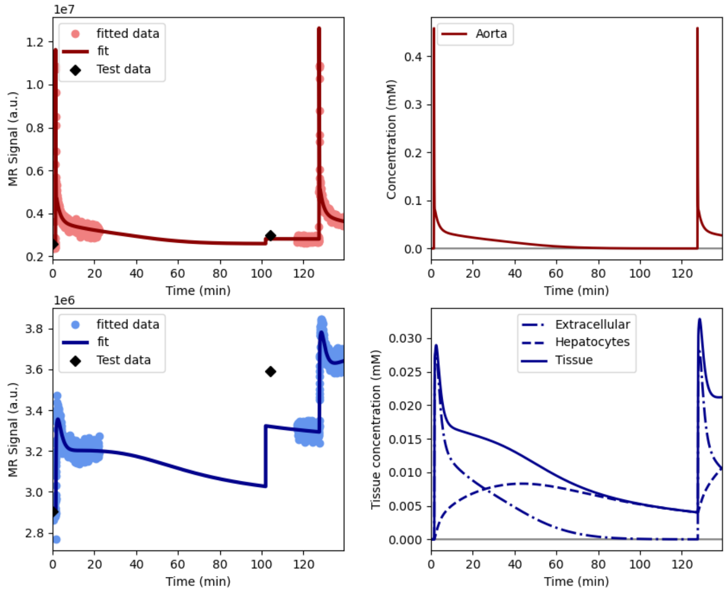

Figure 1.2: Signal-time curves for subject GOT-001 at the control visit.

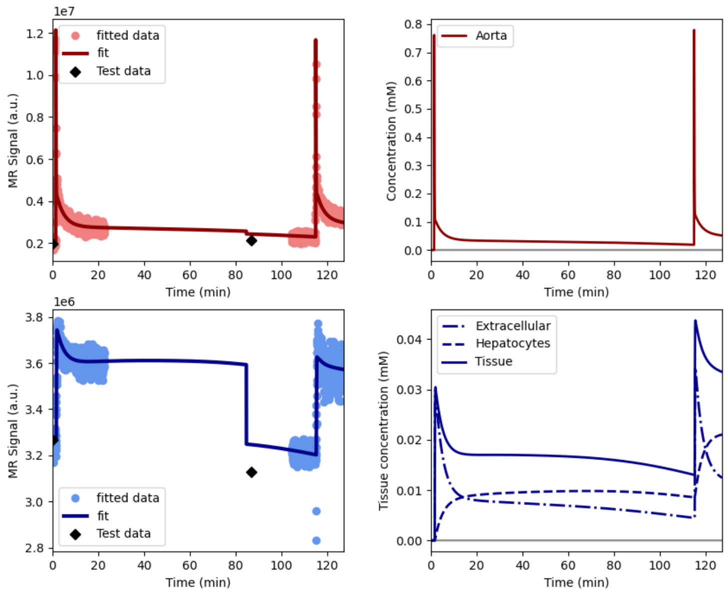

Figure 1.3: Signal-time curves for subject GOT-001 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	37.54	39.44	5.06
AUC for Cl (0-inf)	mM*sec	87.8	164.79	87.69
Biliary excretion rate	mL/min/100cm3	0.92	5.34	481.69
Biliary tissue excretion rate	/min	0.01	0.07	374.39
Final biliary excretion rate	mL/min/100cm3	0.5	5.42	991.17
Final hepatocellular uptake rate	mL/min/100cm3	0.82	1.67	102.4
Gut dispersion	%	83.16	69.15	-16.84
Gut mean transit time	sec	54.57	51.15	-6.27
Hepatocellular mean transit time	min	67.62	14.25	-78.92
Hepatocellular tissue uptake rate	/min	0.04	0.1	140.79
Hepatocellular uptake rate	mL/min/100cm3	0.88	1.34	51.44
Initial biliary excretion rate	mL/min/100cm3	6.17	5.27	-14.44
Initial hepatocellular uptake rate	mL/min/100cm3	0.94	1.01	6.82
Liver T1-MOLLI at 45min	sec	0.85	0.77	-10.15
Liver T1-MOLLI at baseline	sec	0.99	0.9	-8.95
Liver T1-MOLLI at scan 2	sec	0.86	0.84	-2.09
Liver extracellular volume fraction	mL/100cm3	20.83	13.1	-37.11
Liver plasma clearance	L/min	0.01	0.01	59.66
RE for R1l at 20min	%	11.81	12.49	5.78
RE for Sl at 20min	%	11.67	11.04	-5.38

Table 1.6: Values for liver of subject GOT-001

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	56.98	87.19	53.02
AUC for Cb (0-inf)	mM*sec	83.69	300.35	258.89
Body extraction fraction	%	5.12	1.0	-80.42
Cardiac output	L/min	6.94	5.14	-25.9
Heart-lung dispersion	%	41.67	40.88	-1.89
Heart-lung mean transit time	sec	21.86	18.34	-16.1
Organs blood mean transit time	sec	25.61	35.0	36.67
Organs extraction fraction	%	20.44	16.79	-17.84
Organs extravascular mean transit time	min	3.75	10.01	166.59
RE for R1b at 20min	%	26.44	42.0	58.82
RE for Sb at 20min	%	28.1	46.49	65.42

Table 1.7: Values for aorta of subject GOT-001

1.4.2. Subject GOT-002

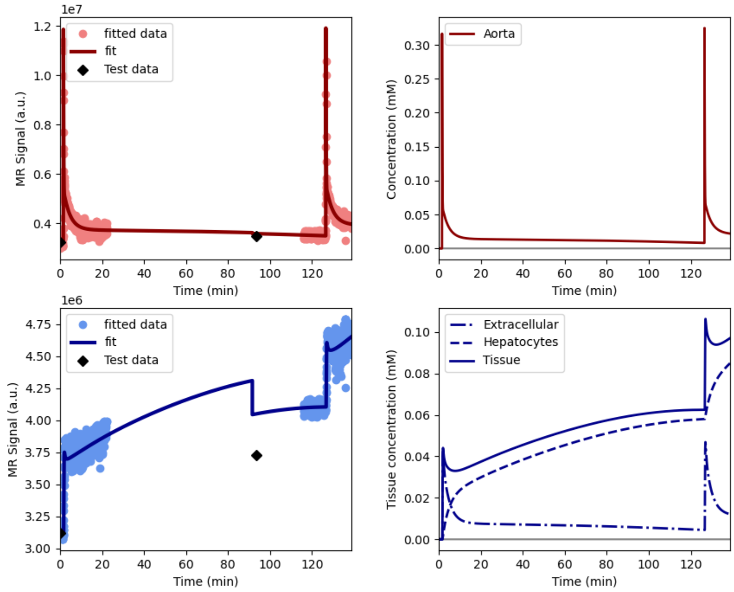

Figure 1.4: Signal-time curves for subject GOT-002 at the control visit.

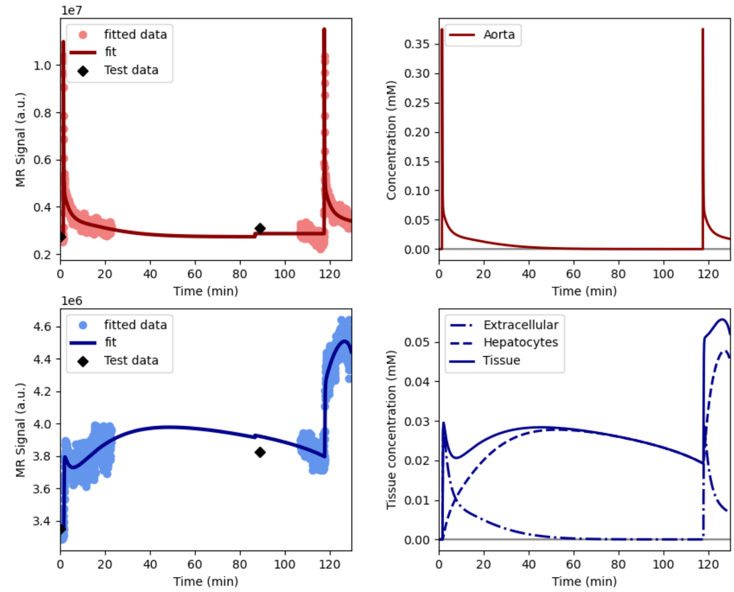

Figure 1.5: Signal-time curves for subject GOT-002 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	82.67	62.19	-24.78
AUC for Cl (0-inf)	mM*sec	582.3	266.94	-54.16
Biliary excretion rate	mL/min/100cm3	0.77	0.22	-71.32
Biliary tissue excretion rate	/min	0.02	0.0	-79.13
Final biliary excretion rate	mL/min/100cm3	0.49	6.24	1186.68
Final hepatocellular uptake rate	mL/min/100cm3	4.6	6.39	38.83
Gut dispersion	%	73.0	73.64	0.88
Gut mean transit time	sec	35.86	51.74	44.29
Hepatocellular mean transit time	min	59.18	283.53	379.06
Hepatocellular tissue uptake rate	/min	0.17	0.2	20.44
Hepatocellular uptake rate	mL/min/100cm3	5.05	4.19	-17.1
Initial biliary excretion rate	mL/min/100cm3	1.84	0.11	-93.91
Initial hepatocellular uptake rate	mL/min/100cm3	5.5	1.98	-63.95
Liver T1-MOLLI at 45min	sec	0.66	0.69	5.95
Liver T1-MOLLI at baseline	sec	0.87	0.84	-3.06
Liver T1-MOLLI at scan 2	sec	0.65	0.72	11.38
Liver extracellular volume fraction	mL/100cm3	30.0	20.65	-31.17
Liver plasma clearance	L/min	0.06	0.06	-8.4
RE for R1l at 20min	%	29.82	19.35	-35.11
RE for Sl at 20min	%	26.31	14.14	-46.23

Table 1.8: Values for liver of subject GOT-002

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	38.16	38.34	0.46
AUC for Cb (0-inf)	mM*sec	133.6	46.31	-65.33
Body extraction fraction	%	1.0	9.28	826.53
Cardiac output	L/min	7.52	5.99	-20.32
Heart-lung dispersion	%	45.29	50.94	12.47
Heart-lung mean transit time	sec	24.0	22.7	-5.42
Organs blood mean transit time	sec	32.31	37.62	16.44
Organs extraction fraction	%	20.96	17.19	-18.01
Organs extravascular mean transit time	min	11.82	5.01	-57.58
RE for R1b at 20min	%	16.47	15.15	-8.01
RE for Sb at 20min	%	14.38	15.14	5.32

Table 1.9: Values for aorta of subject GOT-002

1.4.3. Subject GOT-003

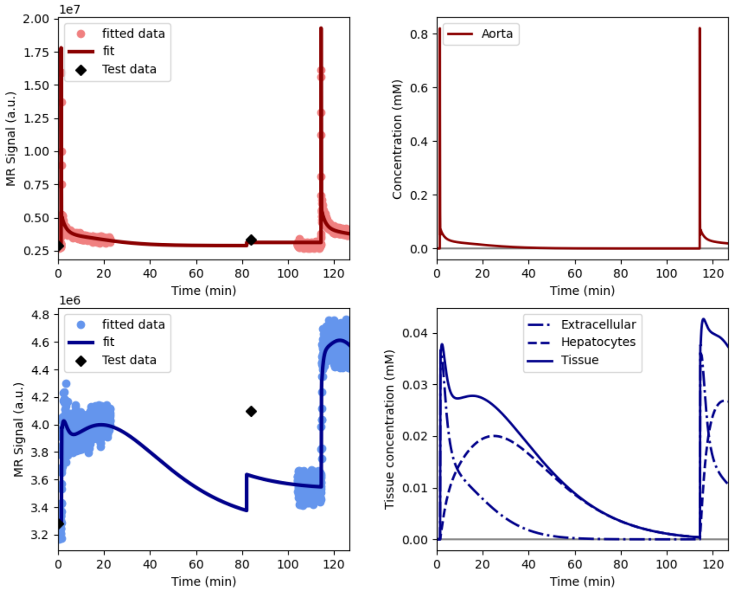

Figure 1.6: Signal-time curves for subject GOT-003 at the control visit.

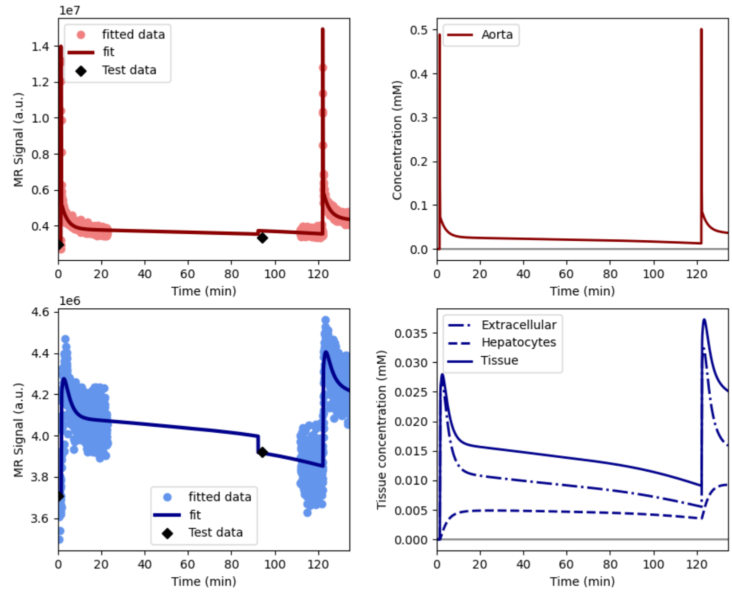

Figure 1.7: Signal-time curves for subject GOT-003 at the treatment visit.

Biomarker	Units	control	drug	change (%)
AUC for Cl (0-35min)	mM*sec	59.73	35.78	-40.1
AUC for Cl (0-inf)	mM*sec	100.17	135.28	35.06
Biliary excretion rate	mL/min/100cm3	2.03	5.75	182.82
Biliary tissue excretion rate	/min	0.04	0.1	123.7
Final biliary excretion rate	mL/min/100cm3	4.55	5.75	26.43
Final hepatocellular uptake rate	mL/min/100cm3	6.88	1.37	-80.03
Gut dispersion	%	87.68	85.42	-2.59
Gut mean transit time	sec	54.97	60.0	9.16
Hepatocellular mean transit time	min	22.37	10.0	-55.3
Hepatocellular tissue uptake rate	/min	0.16	0.05	-67.59
Hepatocellular uptake rate	mL/min/100cm3	4.74	1.2	-74.73
Initial biliary excretion rate	mL/min/100cm3	1.31	5.75	339.21
Initial hepatocellular uptake rate	mL/min/100cm3	2.6	1.02	-60.72
Liver T1-MOLLI at 45min	sec	0.87	0.92	5.19
Liver T1-MOLLI at baseline	sec	1.08	1.0	-7.71
Liver T1-MOLLI at scan 2	sec	0.91	0.92	0.46
Liver extracellular volume fraction	mL/100cm3	30.0	23.39	-22.02
Liver plasma clearance	L/min	0.05	0.01	-74.32
RE for R1l at 20min	%	25.94	11.62	-55.2
RE for Sl at 20min	%	20.12	8.24	-59.08

Table 1.10: Values for liver of subject GOT-003

Biomarker	Units	control	drug	change (%)
AUC for Cb (0-35min)	mM*sec	40.54	62.19	53.4
AUC for Cb (0-inf)	mM*sec	47.84	215.41	350.27
Body extraction fraction	%	7.5	1.0	-86.66
Cardiac output	L/min	6.42	6.1	-5.0
Heart-lung dispersion	%	27.29	39.07	43.15
Heart-lung mean transit time	sec	13.96	19.38	38.79
Organs blood mean transit time	sec	30.56	31.0	1.43
Organs extraction fraction	%	18.02	17.61	-2.27
Organs extravascular mean transit time	min	3.71	7.06	90.21
RE for R1b at 20min	%	17.03	31.01	82.11
RE for Sb at 20min	%	18.65	19.15	2.69

Table 1.11: Values for aorta of subject GOT-003

2

Secondary results

2.1. Diurnal variation

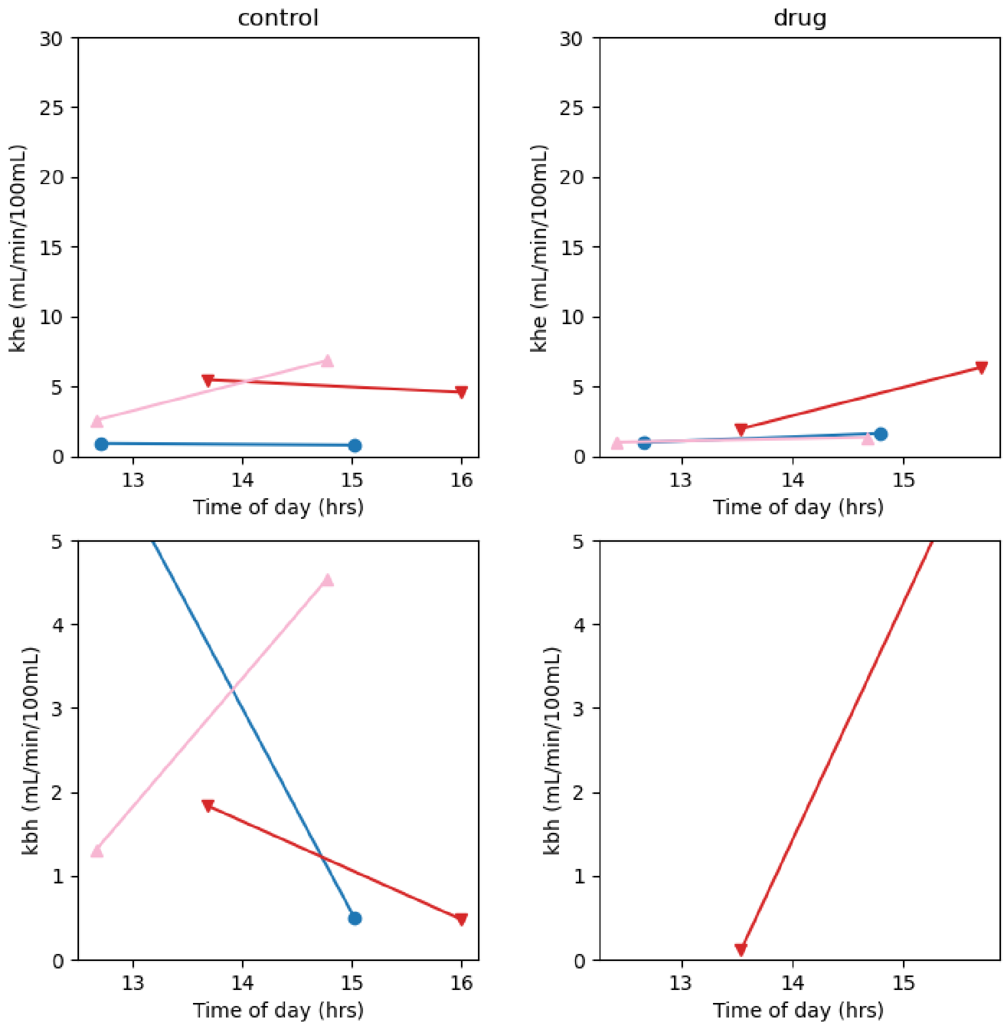

Figure 2.1: Intra-day changes in hepatocellular uptake (k_he, top row) and biliary excretion (k_bh, bottom row) of gadoxetate at for the control (left column) and treatment (right column). Full lines connect values taken in the same subject at the same day.